

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	KAILASH M	Reg. No.:	192524029
Section / Batch:	2025	Date:	28/07/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given scenario and identify the computational problem involved.
- Apply appropriate Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems wherever applicable.
- Clearly justify the chosen solution approach with proper reasoning and analytical interpretation. • Demonstrate decision-making skills by comparing possible solutions and selecting the most suitable approach. • Use diagrams, stepwise illustrations, or algorithmic representations wherever necessary.
- Maintain clarity, logical organization, and proper technical presentation throughout the answers.

Question Type	Focus Area	Questions
Scenario Based Assessment	Analyze real-world scenarios and apply Brute Force techniques for sorting, searching, Closest-Pair, and Convex-Hull problems	Q1

SECTION A – SCENARIO BASED ASSESSMENT

Code of Conduct

I KAILASH M (192524029) certify that this submission is my original work and that I have adhered to the guidelines specified

for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: KAILASH M

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25%	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
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Clarity, Justification & Presentation	10%	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

Marks Evaluation:

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Problem Understanding & Context Analysis	25%				

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Analytical & Decision-Making Skills	20%				
Solution Design & Approach	15%				
Clarity, Justification & Presentation	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q107 — Scenario: Closest Delivery Hubs (Brute Force)

A logistics company computes distances between delivery hubs to find the closest ones using brute-force computation.

(a) Explain how all pairs are evaluated. (b) Derive the time complexity. (c) Discuss scalability concerns.

Scenario Analysis and Problem Identification

The scenario describes a Closest-Pair of Points problem. Each delivery hub can be modelled as a point in a 2-D plane using its geographic coordinates (x, y) — e.g., a local projection of latitude/longitude, or distances in kilometres from a

reference origin. The logistics company wants to know which two hubs are nearest each other — useful for decisions such as merging under-utilised hubs, identifying redundant coverage, planning short-haul shuttle routes between nearby hubs, or flagging hubs that are suspiciously close and could be consolidated to cut cost. Brute-force computation means examining every possible pair of hubs directly, without any spatial shortcuts, and keeping track of the minimum distance found.

(a) How All Pairs Are Evaluated

Let the n delivery hubs be $H_1, H_2, \dots, H_n$, each with coordinates (x_i, y_i) . The brute-force algorithm evaluates every unordered pair (H_i, H_j) , $i \neq j$, exactly once, using two nested loops: the outer loop fixes H_i and the inner loop compares it against every hub H_j that comes after it in the list. For each pair, the Euclidean distance is computed as:

$$d(H_i, H_j) = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$$

A running variable `min_dist` is initialised to infinity (or the first computed distance). As each pair is evaluated, if its distance is smaller than `min_dist`, `min_dist` and the corresponding hub pair are updated. After all $C(n,2) = n(n-1)/2$ pairs have been checked, `min_dist` holds the true minimum distance and its associated pair is the closest pair — the algorithm is exhaustive, so this result is guaranteed to be globally correct, not just locally optimal.

Worked Illustration — 6 Delivery Hubs

Consider 6 hubs with coordinates (in km, relative to a regional origin): $H_1(2,3)$, $H_2(5,8)$, $H_3(9,2)$, $H_4(4,5)$, $H_5(12,7)$, $H_6(6,1)$. There are $C(6,2) = 15$ pairs to evaluate; all 15 distances are computed below.

Pair	Distance (km)
H1–H2	5.831
H1–H3	7.071
H1–H4	2.828 ★ MIN
H1–H5	10.770
H1–H6	4.472
H2–H3	7.211
H2–H4	3.162
H2–H5	7.071
H2–H6	7.071
H3–H4	5.831
H3–H5	5.831
H3–H6	3.162

H4–H5	8.246
H4–H6	4.472
H5–H6	8.485

Scanning all 15 computed distances, the minimum is 2.828 km, between H1(2,3) and H4(4,5) — this is the closest pair, found only after every other pair was also computed and compared, which is the defining trait of the brute-force approach.

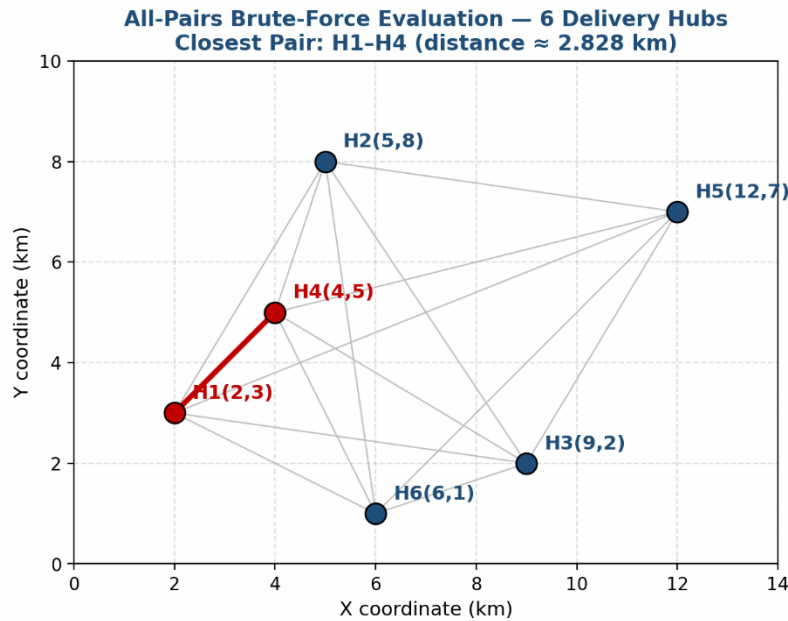


Fig. 1 — All 15 pairwise hub connections (grey); the closest pair H1–H4 is highlighted in red.

(b) Derivation of Time Complexity

The algorithm's cost is dominated by the number of pairwise distance computations. With n hubs, the number of unordered pairs is:

$$C(n,2) = n(n-1) / 2$$

Each pairwise distance computation — two subtractions, two squarings, one addition, one square root — takes $O(1)$ time (constant, independent of n). Multiplying the number of pairs by the constant per-pair cost:

$$T(n) = [n(n-1) / 2] \times O(1) = O(n^2)$$

Unlike search algorithms that can terminate early once a target is found, the closest-pair brute-force method cannot stop early — the global minimum can only be confirmed after every single pair has been examined. This means the best case, average case, and worst case are all identically $O(n^2)$; there is no favourable input that reduces the work below $n(n-1)/2$ comparisons. Space complexity is $O(1)$ additional space beyond storing the n input points (only `min_dist` and the current best pair need to be tracked), i.e. $O(n)$ overall including the input.

For the worked example, $n = 6$ gives $C(6,2) = 15$ distance computations, matching the 15-row table above exactly.

(c) Scalability Concerns

Because the brute-force algorithm's cost grows as $O(n^2)$, its running time increases quadratically as the logistics network

expands — doubling the number of hubs quadruples the number of distance computations, not merely doubles it. This is a serious concern for a logistics company because hub networks are rarely static: national or global carriers routinely operate hundreds to tens of thousands of hubs and micro-depots, and this count grows over time as the business expands.

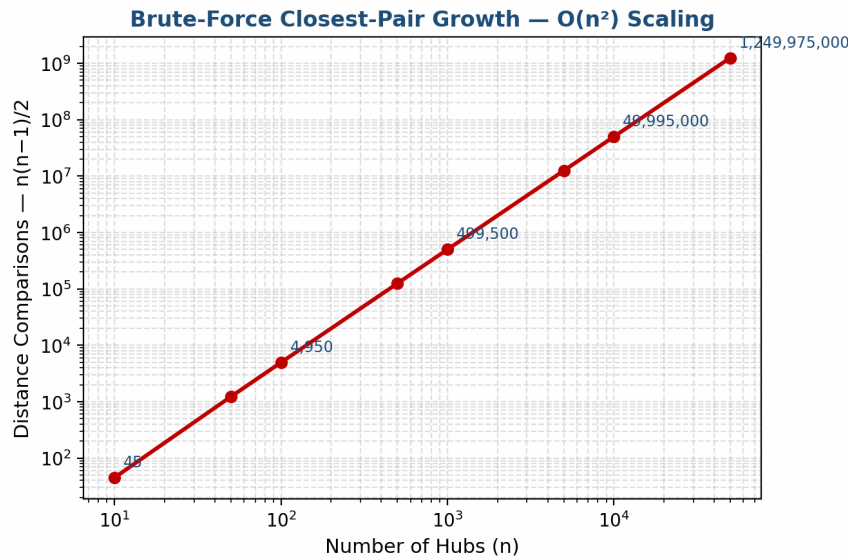


Fig. 2 — Brute-force comparison count grows quadratically; at 50,000 hubs it exceeds 1.2 billion pairwise computations.

- Poor scaling at large n : at $n = 1,000$ hubs the algorithm performs about 500,000 comparisons — still fast; but at $n = 50,000$ hubs (a plausible size for a large national parcel network) it requires over 1.2 billion comparisons, which becomes computationally expensive and slow to run on demand.
- Costly recomputation on updates: if hubs are frequently added, removed, or relocated (common in a growing logistics network), brute force must re-scan effectively all pairs again to guarantee the new global minimum is correct, wasting most of the previous computation — it does not support efficient incremental updates.
- Unsuitability for real-time operations: route-planning and hub-consolidation decisions increasingly need near-real-time answers; an $O(n^2)$ recomputation across a large network is too slow to run interactively or on a live dashboard.
- Memory and infrastructure cost: while brute force itself needs only $O(n)$ memory, the sheer volume of repeated pairwise computations at scale increases CPU time and energy/infrastructure cost when run frequently across a large, growing hub network.

Recommended Mitigation for Scale

For small or moderate hub counts (tens to a few hundred hubs, e.g. a regional distribution network), brute force remains a perfectly reasonable, simple, and exact choice, and its $O(n^2)$ cost is negligible in practice. For large-scale national or global networks, the divide-and-conquer Closest-Pair algorithm reduces the time complexity to $O(n \log n)$ by recursively splitting the hub set, solving each half, and checking only a narrow boundary strip for cross-region closest pairs — a substantial asymptotic improvement. Alternatively, spatial indexing structures such as k-d trees, quad-trees, or

grid-based spatial hashing support efficient nearest-neighbour style queries (typically $O(\log n)$ per query after $O(n \log n)$ preprocessing) and, importantly, support efficient incremental insertion and deletion of hubs — directly addressing the dynamic, frequently-updated nature of a real logistics network. The company should therefore treat brute force as suitable for small batch analyses or as a correctness baseline, while adopting a spatial-indexing or divide-and-conquer approach as the hub network scales into the thousands.

Rubrics for Calculation

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Marks Evaluation

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Faculty In-charge

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Course Coordinator

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Head of Department

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